

Measure Theory and Probability

Lecture 3 — Measurable Functions and the Lebesgue Integral

Benoit Duvocelle

Zhongfa Aviation Institute — Hangzhou

Semester 7 — 2026–2027

Objectives

This lecture first reviews the sigma-algebra and measure tools that will be needed, then develops measurable functions before constructing the Lebesgue integral.

- Review the notation and properties of sigma-algebras and measures used in Chapters 4 and 5.
- Define measurable and Lebesgue-measurable functions, and prove the main stability properties.
- Know how to verify measurability using a generating family and the usual operations.
- Construct the Lebesgue integral first for simple functions, then for non-negative functions and functions of arbitrary sign.
- Master monotone convergence, Fatou's lemma, dominated convergence, and parameter-dependent integrals.

Outline

- 1 Review: sigma-algebras and measures
- 2 Measurable functions
- 3 Overview of the construction
- 4 Simple functions and approximation
- 5 Construction of the Lebesgue integral
- 6 Fundamental properties
- 7 The three convergence theorems
- 8 Parameter-dependent integrals
- 9 Practical methods and summary

Definition

Let Ω be a set. A family $\mathcal{T} \subset \mathcal{P}(\Omega)$ is a **sigma-algebra** if:

- $\emptyset \in \mathcal{T}$;
- $A \in \mathcal{T} \Rightarrow A^c \in \mathcal{T}$;
- $(A_n)_{n \in \mathbb{N}} \subset \mathcal{T} \Rightarrow \bigcup_{n \in \mathbb{N}} A_n \in \mathcal{T}$.

The pair $(\Omega, \mathcal{T})$ is a **measurable space**.

Immediate consequences

$\Omega \in \mathcal{T}$; $\mathcal{T}$ is closed under countable intersections, finite unions/intersections, set differences $A \setminus B$, and symmetric differences $A \triangle B$.

Review 2 — Generated sigma-algebra and Borel sigma-algebra

For $\mathcal{G} \subset \mathcal{P}(\Omega)$, write

$\sigma(\mathcal{G})$ = the smallest sigma-algebra containing $\mathcal{G}$.

On $\mathbb{R}$,

$$\mathcal{B}(\mathbb{R}) = \sigma(\{\text{open subsets of } \mathbb{R}\}).$$

The sigma-algebra $\mathcal{B}(\mathbb{R})$ can also be generated by the half-lines

$$]-\infty, a], \quad a \in \mathbb{R}.$$

More generally, $\mathcal{B}(\mathbb{R}^n)$ is generated by rectangles, for example

$$\prod_{i=1}^n]-\infty, a_i], \quad a_i \in \mathbb{R}.$$

Why this review?

This will allow us to verify measurability on a simple generating family rather than on all Borel sets.

Definition

On a measurable space $(\Omega, \mathcal{T})$, a map

$$\mu : \mathcal{T} \rightarrow \mathbb{R}^+ \cup \{+\infty\}$$

is a **measure** if

- ① $\mu(\emptyset) = 0$;
- ② for every pairwise disjoint family (A_n) ,

$$\mu\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \sum_{n \in \mathbb{N}} \mu(A_n).$$

The triple $(\Omega, \mathcal{T}, \mu)$ is a **measure space**.

Review 4 — Properties of a measure

For $A, B \in \mathcal{T}$:

- **Croissance** : $A \subset B \Rightarrow \mu(A) \leq \mu(B)$.
- **Finite additivity**: if $A \cap B = \emptyset$, then $\mu(A \cup B) = \mu(A) + \mu(B)$.
- **Continuity from below**: if $A_n \uparrow A$, then

$$\mu(A_n) \longrightarrow \mu(A).$$

- **Continuity from above**: if $A_n \downarrow A$ and $\mu(A_m) < \infty$ for some m , then

$$\mu(A_n) \longrightarrow \mu(A).$$

Use in this lecture

Continuity from below of μ is used directly in the proof of monotone convergence for integrals.

Review 5 — Three measures to know

Lebesgue measure λ on $\mathbb{R}$: $\lambda([a, b]) = b - a$; countable sets are null.

Counting measure $\#$: $\#(A) = \text{Card}(A)$, possibly $+\infty$.

Dirac δ_x :

$$\delta_x(A) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Idea

The same Lebesgue integral will yield ordinary integrals, series, or point evaluations depending on the chosen measure.

Review 6 — Null sets and almost everywhere

A set $N \in \mathcal{T}$ is **null** if $\mu(N) = 0$. A property $P(x)$ holds **μ -almost everywhere** (a.e.) if

$$\mu(\{x \in \Omega : P(x) \text{ is false}\}) = 0.$$

Example

For Lebesgue measure,

$$\lambda(\mathbb{Q}) = 0,$$

hence $\mathbb{1}_{\mathbb{Q}} = 0$ almost everywhere.

This notion will be essential for integrability and the convergence theorems.

Preimage: notation

For $f : E \rightarrow F$ and $B \subset F$,

$$f^{-1}(B) = \{x \in E : f(x) \in B\}.$$

This is the **preimage of a set**: the notation is defined even when f is neither injective nor surjective.

Guiding idea

Measurability is entirely based on how the preimage sends measurable sets in the codomain to measurable sets in the domain.

Property 4.2 — Algebraic properties of preimages

For $f : E \rightarrow F$:

- ❶ $f^{-1}(\emptyset) = \emptyset$ et $f^{-1}(F) = E$;
- ❷ $f^{-1}(B^c) = (f^{-1}(B))^c$;
- ❸ for every family $(B_i)_{i \in I}$,

$$f^{-1}\left(\bigcup_{i \in I} B_i\right) = \bigcup_{i \in I} f^{-1}(B_i), \quad f^{-1}\left(\bigcap_{i \in I} B_i\right) = \bigcap_{i \in I} f^{-1}(B_i).$$

Preimages also preserve disjointness.

Proof: complement

Let $B \subset F$ and $x \in E$. Then

$$\begin{aligned}x \in f^{-1}(B^c) &\iff f(x) \in B^c \\&\iff f(x) \notin B \\&\iff x \notin f^{-1}(B) \\&\iff x \in (f^{-1}(B))^c.\end{aligned}$$

Since this equivalence holds for every $x \in E$,

$$f^{-1}(B^c) = (f^{-1}(B))^c.$$

Proof: unions and intersections

For unions,

$$\begin{aligned}x \in f^{-1}\left(\bigcup_{i \in I} B_i\right) &\iff f(x) \in \bigcup_{i \in I} B_i \\&\iff \exists i \in I, f(x) \in B_i \\&\iff \exists i \in I, x \in f^{-1}(B_i) \\&\iff x \in \bigcup_{i \in I} f^{-1}(B_i).\end{aligned}$$

Similarly,

$$x \in f^{-1}\left(\bigcap_{i \in I} B_i\right) \iff \forall i, x \in f^{-1}(B_i) \iff x \in \bigcap_{i \in I} f^{-1}(B_i).$$

Finally, if $B_i \cap B_j = \emptyset$ and $x \in f^{-1}(B_i) \cap f^{-1}(B_j)$, then $f(x) \in B_i \cap B_j = \emptyset$, a contradiction.

$f^{-1}(\mathcal{S})$ is a sigma-algebra

Property

If $(F, \mathcal{S})$ is a measurable space and $f : E \rightarrow F$, then

$$f^{-1}(\mathcal{S}) := \{f^{-1}(B) : B \in \mathcal{S}\}$$

is a sigma-algebra on E .

Proof.

- $f^{-1}(\emptyset) = \emptyset$;
- if $A = f^{-1}(B)$, then

$$A^c = f^{-1}(B^c) \in f^{-1}(\mathcal{S});$$

- if $A_n = f^{-1}(B_n)$,

$$\bigcup_n A_n = f^{-1}\left(\bigcup_n B_n\right) \in f^{-1}(\mathcal{S}).$$

The three sigma-algebra axioms are satisfied.

Property

For $f : E \rightarrow F$ and $\mathcal{G} \subset \mathcal{P}(F)$,

$$f^{-1}(\sigma(\mathcal{G})) = \sigma(f^{-1}(\mathcal{G})).$$

First inclusion. $f^{-1}(\sigma(\mathcal{G}))$ is a sigma-algebra containing $f^{-1}(\mathcal{G})$; therefore it contains the smallest sigma-algebra generated by $f^{-1}(\mathcal{G})$:

$$\sigma(f^{-1}(\mathcal{G})) \subset f^{-1}(\sigma(\mathcal{G})).$$

Generated sigma-algebra and preimage: second inclusion

Set

$$\mathcal{S} = \{B \subset F : f^{-1}(B) \in \sigma(f^{-1}(\mathcal{G}))\}.$$

By the algebraic properties of f^{-1} , $\mathcal{S}$ is a sigma-algebra:

- $\emptyset \in \mathcal{S}$;
- $B \in \mathcal{S} \Rightarrow B^c \in \mathcal{S}$;
- $(B_n) \subset \mathcal{S} \Rightarrow \bigcup_n B_n \in \mathcal{S}$.

Moreover, $\mathcal{G} \subset \mathcal{S}$, hence $\sigma(\mathcal{G}) \subset \mathcal{S}$. Therefore, for every $B \in \sigma(\mathcal{G})$,

$$f^{-1}(B) \in \sigma(f^{-1}(\mathcal{G})),$$

hence

$$f^{-1}(\sigma(\mathcal{G})) \subset \sigma(f^{-1}(\mathcal{G})).$$

The two inclusions give the equality.

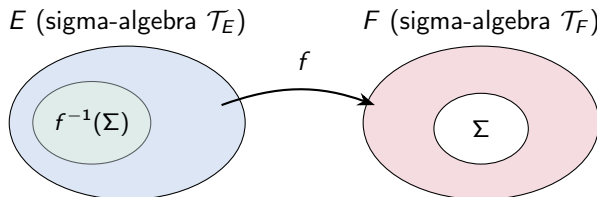
Definition of a measurable function

Definition

Let $(E, \mathcal{T}_E)$ and $(F, \mathcal{T}_F)$ be measurable spaces. A map $f : E \rightarrow F$ is **measurable** if

$$\forall \Sigma \in \mathcal{T}_F, \quad f^{-1}(\Sigma) \in \mathcal{T}_E.$$

If $\mathcal{T}_E$ and $\mathcal{T}_F$ are the Borel sigma-algebras, f is also said to be **Borel measurable**.



Probabilistic motivation

If $X : (\Omega, \mathcal{T}, \mathbb{P}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is a random variable, measurability ensures that events described in terms of its values belong to $\mathcal{T}$.

Example: the event $\{X = 2\}$

Since $\{2\} \in \mathcal{B}(\mathbb{R})$,

$$\{X = 2\} = \{\omega \in \Omega : X(\omega) = 2\} = X^{-1}(\{2\}) \in \mathcal{T}.$$

It can therefore be assigned a probability:

$$\mathbb{P}(X = 2) = \mathbb{P}(X^{-1}(\{2\})).$$

Similarly, for every $a \in \mathbb{R}$,

$$\{X \leq a\} = X^{-1}(]-\infty, a]) \in \mathcal{T},$$

which explains why quantities such as $\mathbb{P}(X \leq a)$ are well defined.

Lebesgue-measurable function

Definition

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is **Lebesgue measurable** if it is measurable from

$$(\mathbb{R}, \mathcal{M}(\lambda^*)) \text{ to } (\mathbb{R}, \mathcal{B}(\mathbb{R})),$$

that is,

$$\forall B \in \mathcal{B}(\mathbb{R}), \quad f^{-1}(B) \in \mathcal{M}(\lambda^*).$$

Every Borel measurable function $f : \mathbb{R} \rightarrow \mathbb{R}$ is Lebesgue measurable because

$$\mathcal{B}(\mathbb{R}) \subseteq \mathcal{M}(\lambda^*).$$

Key point

The sigma-algebra on the domain matters: “Borel measurable” and “Lebesgue measurable” are not exactly the same notion.

Property

If $f : (E, \mathcal{T}_E) \rightarrow (F, \mathcal{T}_F)$ and $g : (F, \mathcal{T}_F) \rightarrow (G, \mathcal{T}_G)$ are measurable, then $g \circ f$ is measurable.

Proof. For $W \in \mathcal{T}_G$,

$$(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W)).$$

Since g is measurable, $g^{-1}(W) \in \mathcal{T}_F$; since f is measurable,

$$f^{-1}(g^{-1}(W)) \in \mathcal{T}_E.$$

Therefore $g \circ f$ is measurable.

Warning: composition of Lebesgue-measurable functions

The previous property assumes that the outer function g is measurable with respect to the sigma-algebra on the intermediate space. A Lebesgue-measurable function

$f : \mathbb{R} \rightarrow \mathbb{R}$ only satisfies

$$f^{-1}(\mathcal{B}(\mathbb{R})) \subseteq \mathcal{M}(\lambda^*),$$

and not necessarily

$$f^{-1}(\mathcal{M}(\lambda^*)) \subseteq \mathcal{M}(\lambda^*).$$

Thus, two functions that are merely Lebesgue measurable may have a composition that is not Lebesgue measurable.

Tutorial 3

The full counterexample is studied in Part 3: “Composition of Lebesgue-measurable functions: a counterexample”.

Characterization by a generating family

Property

Let $\mathcal{G} \subset \mathcal{T}_F$ satisfy $\sigma(\mathcal{G}) = \mathcal{T}_F$. Then $f : E \rightarrow F$ is measurable if and only if

$$f^{-1}(G) \in \mathcal{T}_E \quad \forall G \in \mathcal{G}.$$

The forward implication is immediate. Conversely, the assumption gives

$f^{-1}(\mathcal{G}) \subset \mathcal{T}_E$, hence

$$\sigma(f^{-1}(\mathcal{G})) \subset \mathcal{T}_E.$$

But

$$\sigma(f^{-1}(\mathcal{G})) = f^{-1}(\sigma(\mathcal{G})) = f^{-1}(\mathcal{T}_F).$$

Therefore $f^{-1}(\mathcal{T}_F) \subset \mathcal{T}_E$: f is measurable.

Practical method: proving that a function is measurable

For $f : E \rightarrow \mathbb{R}$, it is often enough to verify

$$\{f \leq a\} = f^{-1}(]-\infty, a]) \in \mathcal{T}_E \quad \forall a \in \mathbb{R}.$$

Indeed, the half-lines $] - \infty, a]$ generate $\mathcal{B}(\mathbb{R})$. In practice, we mainly use the following toolkit:

- continuity $\Rightarrow$ measurability;
- stability under $+$, $\times$, $|\cdot|$;
- stability under countable $\sup$, $\inf$, $\limsup$, $\liminf$;
- pointwise limit of measurable functions $\Rightarrow$ measurable.

Continuous functions are measurable

Property

Between normed vector spaces equipped with their Borel sigma-algebras, every continuous map is measurable.

Proof. The open sets of F generate $\mathcal{B}(F)$. For every open set $V \subset F$, continuity gives

$$f^{-1}(V) \text{ open in } E,$$

hence $f^{-1}(V) \in \mathcal{B}(E)$. The characterization by a generating family gives the result.

Examples

Constant functions, polynomials, exp, sin, cos, etc.

Indicator functions are measurable

Property

If $A \in \mathcal{T}$, then $\mathbb{1}_A : E \rightarrow \mathbb{R}$ is measurable.

Proof. For $B \subset \mathbb{R}$, depending on whether 0 and 1 belong to B ,

$$\mathbb{1}_A^{-1}(B) \in \{\emptyset, A, A^c, E\} \subset \mathcal{T}.$$

Two examples

$\mathbb{1}_{\mathbb{Q}}$ is Borel measurable although it is continuous nowhere. By contrast, if V is a Vitali set, $\mathbb{1}_V$ is not Lebesgue measurable because

$$\{\mathbb{1}_V > 1/2\} = V \notin \mathcal{M}(\lambda^*).$$

Property

For $f = (f_1, \dots, f_n) : E \rightarrow \mathbb{R}^n$,

$$f \text{ measurable} \iff f_1, \dots, f_n \text{ measurable.}$$

Forward implication. Each projection $\pi_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous, hence measurable; therefore

$$f_i = \pi_i \circ f$$

is measurable by composition.

Measurable functions with values in $\mathbb{R}^n$: converse

Assume $f_1, \dots, f_n$ are measurable. Rectangles $B_1 \times \dots \times B_n$, $B_i \in \mathcal{B}(\mathbb{R})$, generate $\mathcal{B}(\mathbb{R}^n)$ and

$$f^{-1}(B_1 \times \dots \times B_n) = \bigcap_{i=1}^n f_i^{-1}(B_i).$$

Each $f_i^{-1}(B_i) \in \mathcal{T}$; therefore their finite intersection belongs to $\mathcal{T}$. By the characterization using a generating family, f is measurable.

Consequence

For two real-valued measurable functions f, g , the map

$$h = (f, g) : E \rightarrow \mathbb{R}^2$$

is measurable.

Stability under the usual operations

Property

If $f, g : E \rightarrow \mathbb{R}$ are measurable and $\lambda \in \mathbb{R}$, then

$$f + g, \quad fg, \quad \lambda f, \quad |f|$$

are measurable.

Proof. $h = (f, g) : E \rightarrow \mathbb{R}^2$ is measurable. The continuous maps

$$S(x, y) = x + y, \quad P(x, y) = xy, \quad m_\lambda(x) = \lambda x, \quad \text{abs}(x) = |x|$$

are measurable. By composition,

$$S \circ h = f + g, \quad P \circ h = fg, \quad m_\lambda \circ f = \lambda f, \quad \text{abs} \circ f = |f|.$$

The extended real line

Set

$$\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}.$$

Equip $\overline{\mathbb{R}}$ with the sigma-algebra

$$\mathcal{B}(\overline{\mathbb{R}}) = \{A \subset \overline{\mathbb{R}} : A \cap \mathbb{R} \in \mathcal{B}(\mathbb{R})\}.$$

It is generated, for example, by

$$[-\infty, a], \quad a \in \mathbb{R}.$$

Important convention

We systematically avoid $\infty - \infty$. In particular, we use the convention $0 \times \infty = 0$.

Countable supremum and infimum

Property

If $(f_n)_{n \geq 1}$ are measurable from E to $\overline{\mathbb{R}}$, then

$$\sup_{n \geq 1} f_n \quad \text{et} \quad \inf_{n \geq 1} f_n$$

are measurable.

Proof for the supremum. For $a \in \mathbb{R}$,

$$\begin{aligned} \left\{ \sup_n f_n \leq a \right\} &= \bigcap_{n \geq 1} \{f_n \leq a\} \\ &= \bigcap_{n \geq 1} f_n^{-1}([-\infty, a]) \in \mathcal{T}. \end{aligned}$$

The half-line criterion gives the result. For the infimum, use $[a, +\infty]$ symmetrically.

Limsup, liminf, and pointwise limits

Define

$$\limsup_n f_n = \inf_{n \geq 1} \sup_{m \geq n} f_m, \quad \liminf_n f_n = \sup_{n \geq 1} \inf_{m \geq n} f_m.$$

By stability under countable suprema and infima, both functions are measurable. If $f_n(x) \rightarrow f(x)$ for every x , then

$$f = \limsup_n f_n = \liminf_n f_n,$$

hence

$f_n \text{ measurable and } f_n \rightarrow f \text{ pointwise} \Rightarrow f \text{ measurable.}$

Further topic: pushforward measure

Definition

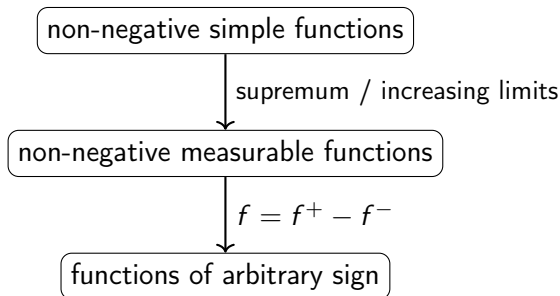
If $f : (E, \mathcal{T}_E) \rightarrow (F, \mathcal{T}_F)$ is measurable and μ is a measure on $(E, \mathcal{T}_E)$, the **pushforward measure** of μ by f is

$$f(\mu)(B) = \mu(f^{-1}(B)), \quad B \in \mathcal{T}_F.$$

It is also denoted by μ_f .

In probability, for a random variable X , this construction is the **distribution** $\mathbb{P}_X$ of X .

The construction in three stages



Guiding idea

Measure level sets rather than partitioning the domain into intervals.

Definition

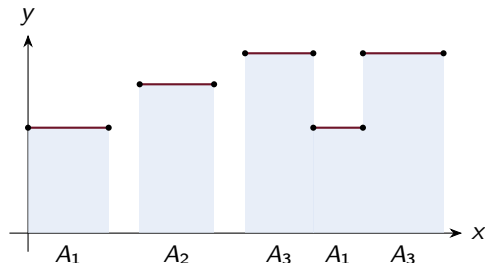
Let $(E, \mathcal{T})$ be a measurable space. A function $f : E \rightarrow \mathbb{R}$ is **simple** if there exist pairwise disjoint measurable sets $A_1, \dots, A_p \in \mathcal{T}$ and real numbers $x_1, \dots, x_p$ such that

$$f = \sum_{k=1}^p x_k \mathbb{1}_{A_k}.$$

Measurability

Every simple function is measurable: each $\mathbb{1}_{A_k}$ is measurable and a finite linear combination of measurable functions is measurable.

Graph: a simple function



The sets A_k are only assumed to be measurable: they need not be intervals. Here A_1 and A_3 each consist of two disjoint pieces on which the function takes the same value.

Theorem 5.4 — Approximation lemma

Theorem 5.4 — Approximation lemma

If $f : E \rightarrow [0, +\infty]$ is measurable, there exists a sequence of non-negative simple functions (f_n) such that

$$0 \leq f_1 \leq f_2 \leq \cdots \leq f, \quad f_n(x) \longrightarrow f(x) \quad (x \in E).$$

A canonical choice is

$$f_n(x) = \min(n, 2^{-n} \lfloor 2^n f(x) \rfloor).$$

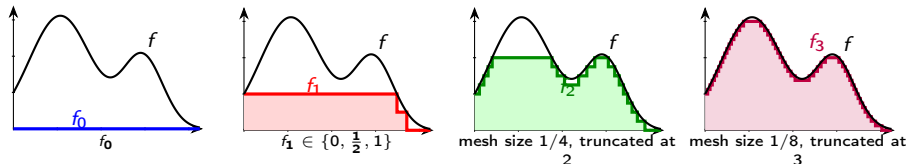


Figure from the course notes: $f_0 \leq f_1 \leq f_2 \leq f_3 \leq f$.

Proof of Theorem 5.4 — Construction

For $n \geq 1$ and $0 \leq k < n2^n$, set

$$A_{n,k} = \left\{ x \in E : \frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n} \right\} = f^{-1} \left(\left[\frac{k}{2^n}, \frac{k+1}{2^n} \right) \right),$$

and

$$A_n = \{x \in E : f(x) \geq n\} = f^{-1}([n, +\infty)).$$

These sets are measurable, pairwise disjoint, and

$$E = A_n \cup \bigcup_{k=0}^{n2^n-1} A_{n,k}.$$

Then define

$$f_n = \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \mathbb{1}_{A_{n,k}} + n \mathbb{1}_{A_n}.$$

Thus f_n is simple, non-negative, and $f_n \leq f$.

Proof of Theorem 5.4 — Monotonicity (1/2)

Fix $x \in E$.

Cas 1 : $f(x) \geq n + 1$.

$$f_n(x) = n < n + 1 = f_{n+1}(x).$$

Cas 2 : $n \leq f(x) < n + 1$. Then $f_n(x) = n$. For some integer k' ,

$$f_{n+1}(x) = \frac{k'}{2^{n+1}}, \quad k' \geq n2^{n+1},$$

hence

$$f_n(x) = n \leq \frac{k'}{2^{n+1}} = f_{n+1}(x).$$

Proof of Theorem 5.4 — Monotonicity (2/2)

Cas 3 : $0 \leq f(x) < n$. There exists $k \in \{0, \dots, n2^n - 1\}$ such that

$$\frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n}.$$

Split this interval into two halves.

If

$$\frac{2k}{2^{n+1}} \leq f(x) < \frac{2k+1}{2^{n+1}},$$

then

$$f_n(x) = \frac{k}{2^n} = \frac{2k}{2^{n+1}} = f_{n+1}(x).$$

If

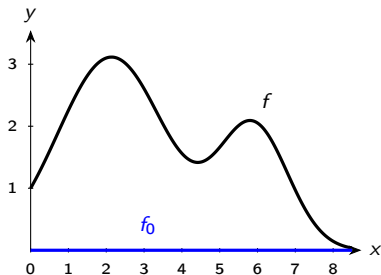
$$\frac{2k+1}{2^{n+1}} \leq f(x) < \frac{2k+2}{2^{n+1}},$$

then

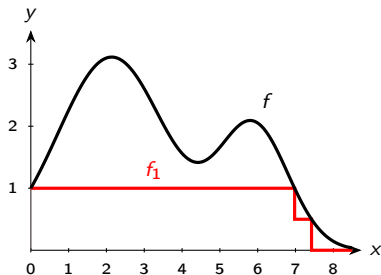
$$f_n(x) = \frac{2k}{2^{n+1}} < \frac{2k+1}{2^{n+1}} = f_{n+1}(x).$$

Thus, in every case, $f_n(x) \leq f_{n+1}(x)$: the sequence (f_n) is increasing.

Approximation by simple functions: f_0 and f_1

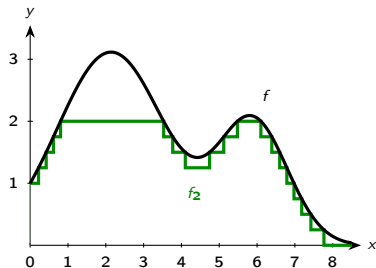


$f_0(x) = 0$ for every x .

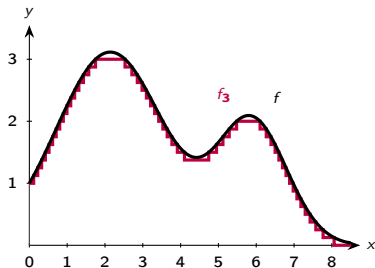


$f_1 \in \{0, \frac{1}{2}, 1\}$, the largest function below f with these values.

Approximation by simple functions: f_2 and f_3



f_2 : mesh size 1/4, truncated at 2.



f_3 : mesh size 1/8, truncated at 3.

Proof of Theorem 5.4 — Pointwise convergence

Fix $x \in E$.

If $f(x) = +\infty$, then

$$f_n(x) = n \xrightarrow{n \rightarrow \infty} +\infty = f(x).$$

If $f(x) < +\infty$, take $n > f(x)$. Then truncation no longer applies and there exists k such that

$$\frac{k}{2^n} \leq f(x) < \frac{k+1}{2^n}, \quad f_n(x) = \frac{k}{2^n}.$$

Therefore,

$$0 \leq f(x) - f_n(x) < \frac{k+1}{2^n} - \frac{k}{2^n} = 2^{-n} \xrightarrow{n \rightarrow \infty} 0.$$

Hence $f_n(x) \rightarrow f(x)$ for every $x \in E$.

Conclusion

(f_n) is an increasing sequence of non-negative simple functions converging pointwise to f .

Integral of a non-negative simple function

Let

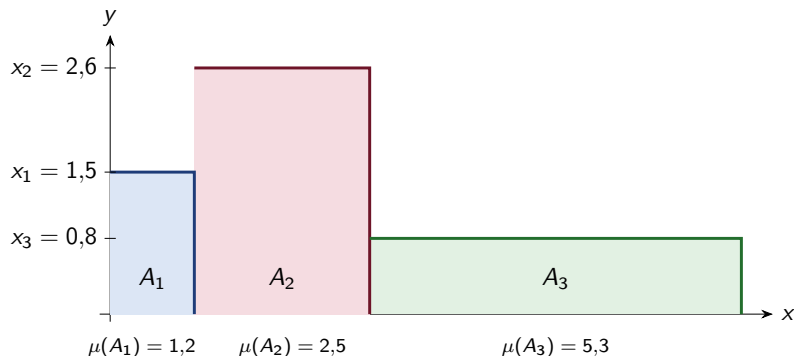
$$S = \sum_{k=1}^p x_k \mathbb{1}_{A_k}, \quad x_k \geq 0,$$

where (A_k) is a measurable partition. Define

$$\int_E S d\mu = \sum_{k=1}^p x_k \mu(A_k).$$

This value does not depend on the chosen representation: use the common refinement $A_i \cap B_j$.

Graphical interpretation of the integral of a simple function



$$\int f \, d\mu = 1,5 \mu(A_1) + 2,6 \mu(A_2) + 0,8 \mu(A_3).$$

The integral is the sum of the “generalized areas” $x_k \mu(A_k)$.

Example 5.10 — Lebesgue and counting measures

Lebesgue measure. On $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$, let

$$f = 2\mathbb{1}_{[0,1]} + 3\mathbb{1}_{\mathbb{Q} \cap [1,4]}.$$

Since $\mathbb{Q} \cap [1, 4]$ is countable,

$$\int_{\mathbb{R}} f \, d\lambda = 2\lambda([0, 1]) + 5\lambda(\mathbb{Q} \cap [1, 4]) = 2.$$

Counting measure. On $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$, for

$$g = 2\mathbb{1}_{\{0,1,2\}} + 4\mathbb{1}_{\{5\}},$$

we obtain

$$\int_{\mathbb{N}} g \, d\# = 2 \times 3 + 4 \times 1 = 10.$$

More generally, for $g \geq 0$,

$$\int_{\mathbb{N}} g \, d\# = \sum_{n \geq 0} g(n).$$

Example 5.10 — Dirac measure at a point

Fix $a \in \mathbb{R}$ and consider δ_a . For

$$h = \sum_{k=1}^p x_k \mathbb{1}_{A_k},$$

where (A_k) is a measurable partition, there is a unique j such that $a \in A_j$. Hence

$$\int_{\mathbb{R}} h d\delta_a = \sum_{k=1}^p x_k \delta_a(A_k) = x_j = \boxed{h(a)}.$$

Concrete calculation

For $a = 2$ and

$$h = 3\mathbb{1}_{[0,1]} + 7\mathbb{1}_{[1,3]},$$

we have

$$\int_{\mathbb{R}} h d\delta_2 = h(2) = 7.$$

Integral of a non-negative measurable function

For measurable $f : E \rightarrow [0, +\infty]$,

$$\int_E f \, d\mu = \sup \left\{ \int_E S \, d\mu : 0 \leq S \leq f, S \text{ simple} \right\}.$$

The value belongs to $[0, +\infty]$.

Positive monotonicity — two supremum steps

Assume $0 \leq u \leq v$.

1. Fix $\varphi \leq u$. If φ is a non-negative simple function, then $\varphi \leq v$, hence

$$\int_E \varphi \, d\mu \leq \sup_{\substack{0 \leq \psi \leq v \\ \psi \text{ simple}}} \int_E \psi \, d\mu = \int_E v \, d\mu.$$

2. Then take the supremum over $\varphi \leq u$.

$$\int_E u \, d\mu = \sup_{\substack{0 \leq \varphi \leq u \\ \varphi \text{ simple}}} \int_E \varphi \, d\mu \leq \int_E v \, d\mu.$$

Conclusion

$$0 \leq u \leq v \implies \int_E u \, d\mu \leq \int_E v \, d\mu.$$

Lemma — Monotonicity for non-negative simple functions

Lemma — Monotonicity for non-negative simple functions

If φ and ψ are two non-negative simple functions such that $0 \leq \varphi \leq \psi$, then

$$\int_E \varphi d\mu \leq \int_E \psi d\mu.$$

Proof. Write

$$\varphi = \sum_{i=1}^p a_i \mathbb{1}_{A_i}, \quad \psi = \sum_{j=1}^q b_j \mathbb{1}_{B_j}.$$

On the common partition $(A_i \cap B_j)_{i,j}$, whenever $A_i \cap B_j \neq \emptyset$, the pointwise inequality gives $a_i \leq b_j$. Hence, term by term,

$$a_i \mu(A_i \cap B_j) \leq b_j \mu(A_i \cap B_j).$$

Summing gives

$$\int_E \varphi d\mu = \sum_{i,j} a_i \mu(A_i \cap B_j) \leq \sum_{i,j} b_j \mu(A_i \cap B_j) = \int_E \psi d\mu.$$

Proposition 5.13 — Increasing limit and integral

Proposition 5.13 — Increasing limit and integral

Let (f_n) be an increasing sequence of non-negative simple functions such that $f_n \rightarrow f$ pointwise. Then

$$\lim_{n \rightarrow \infty} \int_E f_n d\mu = \int_E f d\mu.$$

By the previous lemma,

$$f_n \leq f_{n+1} \implies \int_E f_n d\mu \leq \int_E f_{n+1} d\mu.$$

Therefore

$$L := \lim_n \int_E f_n d\mu$$

exists in $[0, +\infty]$. Since $f_n \leq f$ and f_n is simple,

$$\int_E f_n d\mu \leq \int_E f d\mu, \quad L \leq \int_E f d\mu.$$

Proof of Proposition 5.13 — Reverse inequality

Fix a non-negative simple function $\varphi \leq f$ and $\alpha \in]0, 1[$. Set

$$E_n = \{f_n \geq \alpha\varphi\}.$$

Since $f_n \uparrow f$, we have $E_n \uparrow E$. Moreover,

$$0 \leq \alpha\varphi\mathbb{1}_{E_n} \leq f_n,$$

hence, by the monotonicity lemma for simple functions,

$$\alpha \int_E \varphi \mathbb{1}_{E_n} d\mu \leq \int_E f_n d\mu \leq L.$$

If $\varphi = \sum_{j=1}^p a_j \mathbb{1}_{A_j}$, then

$$\int_E \varphi \mathbb{1}_{E_n} d\mu = \sum_{j=1}^p a_j \mu(A_j \cap E_n) \longrightarrow \sum_{j=1}^p a_j \mu(A_j) = \int_E \varphi d\mu.$$

Thus

$$L \geq \alpha \int_E \varphi d\mu, \quad \alpha \uparrow 1 \implies L \geq \int_E \varphi d\mu.$$

Finally, taking the supremum over all non-negative simple functions $\varphi \leq f$,

$$L \geq \sup_{\substack{0 \leq \varphi \leq f \\ \varphi \text{ simple}}} \int_E \varphi d\mu = \int_E f d\mu.$$

Together with the opposite inequality: $L = \int_E f d\mu$.

Positive and negative parts

For a measurable function $f : E \rightarrow \overline{\mathbb{R}}$, define

$$f^+ = \max(f, 0), \quad f^- = \max(-f, 0).$$

Then

$$f = f^+ - f^-, \quad |f| = f^+ + f^-, \quad f^+ f^- = 0.$$

Both f^+ and f^- are non-negative and measurable.

Integral of a function of arbitrary sign

If at least one of $\int f^+ d\mu$ and $\int f^- d\mu$ is finite, define

$$\int_E f d\mu = \int_E f^+ d\mu - \int_E f^- d\mu.$$

If both are $+\infty$, the expression $+\infty - \infty$ is undefined.

Definition

A real-valued measurable function f is **integrable** if

$$\int_E |f| d\mu < +\infty.$$

Equivalently,

$$\int f^+ d\mu < \infty \quad \text{et} \quad \int f^- d\mu < \infty.$$

Then $\int f d\mu \in \mathbb{R}$.

Integral over a measurable set

For $A \in \mathcal{T}$,

$$\int_A f \, d\mu := \int_E f \mathbb{1}_A \, d\mu.$$

Integrating over a set therefore amounts to multiplying the function by the indicator of that set.

Almost everywhere

A property holds **almost everywhere** (a.e.) if it fails only on a measurable null set.

Example

On $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$,

$$1_{\mathbb{Q}} = 0 \quad \text{a.e.}$$

because $\lambda(\mathbb{Q}) = 0$.

Compatibility of the Riemann and Lebesgue integrals

If $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable, then it is Lebesgue integrable and

$$\int_a^b f(x) dx = \int_{[a,b]} f d\lambda.$$

By contrast, $\mathbb{1}_{\mathbb{Q} \cap [0,1]}$ is not Riemann integrable, whereas

$$\int_{[0,1]} \mathbb{1}_{\mathbb{Q}} d\lambda = 0.$$

Property — Linearity of the integral

If f, g are integrable and $a, b \in \mathbb{R}$, then

$$\int (af + bg) d\mu = a \int f d\mu + b \int g d\mu.$$

For non-negative functions, the analogous formula remains valid in $[0, +\infty]$ whenever no indeterminate expression arises.

Property — Monotonicity of the integral

If $f \leq g$ a.e. and the integrals are defined, then

$$\int f \, d\mu \leq \int g \, d\mu.$$

For $0 \leq f \leq g$, this property was proved directly from the definition of the integral, using two successive supremum steps.

For integrable functions of arbitrary sign, apply it to the positive and negative parts.

In particular,

$$\left| \int f \, d\mu \right| \leq \int |f| \, d\mu.$$

Additivity with respect to sets

Property — Additivity with respect to sets

If A, B are disjoint measurable sets and $f \geq 0$ is measurable, then

$$\int_{A \cup B} f \, d\mu = \int_A f \, d\mu + \int_B f \, d\mu.$$

More generally, for pairwise disjoint (A_n) ,

$$\int_{\bigcup_n A_n} f \, d\mu = \sum_n \int_{A_n} f \, d\mu.$$

Property — Characterization of a zero integral

For measurable $f \geq 0$,

$$\int f \, d\mu = 0 \iff f = 0 \quad \mu\text{-a.e.}$$

Consequently, changing an integrable function on a null set does not change its integral.

Equality almost everywhere

Property — Invariance under equality almost everywhere

If $f = g$ a.e. and one of the two functions is integrable, then the other is also integrable and

$$\int f \, d\mu = \int g \, d\mu.$$

Lebesgue integration naturally identifies functions that differ only on a null set.

Properties — Useful integrability criteria

If $|f| \leq g$ a.e. with $g \in L^1(\mu)$, then $f \in L^1(\mu)$.

If $\mu(E) < \infty$ and f is measurable and bounded, then

$$\int_E |f| d\mu \leq \|f\|_\infty \mu(E) < \infty.$$

Why the convergence theorems are essential

The central question is:

$$f_n(x) \rightarrow f(x) \implies \int f_n d\mu \rightarrow \int f d\mu ?$$

Answer

Not in general. Additional assumptions are needed.

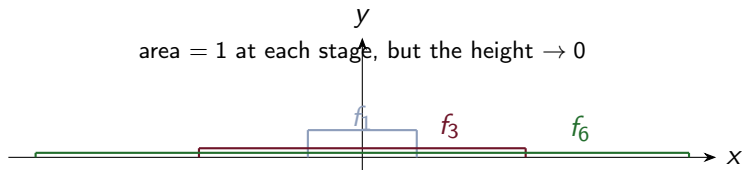
Pointwise convergence: a counterexample

On $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$, set

$$f_n = \frac{1}{2n} 1_{[-n, n]}.$$

Then, for every $n \geq 1$,

$$\int_{\mathbb{R}} f_n d\lambda = \frac{1}{2n} \lambda([-n, n]) = 1, \quad f_n(x) \xrightarrow{n \rightarrow \infty} 0 \quad \text{for every } x \in \mathbb{R}.$$



$$\lim_{n \rightarrow \infty} \int f_n d\lambda = 1 \neq 0 = \int \lim_{n \rightarrow \infty} f_n d\lambda.$$

Monotone convergence theorem

Monotone convergence theorem (Beppo–Levi)

Let (f_n) be a sequence of measurable functions from E to $[0, +\infty]$ such that

$$f_n \uparrow f \quad \text{pointwise.}$$

Then

$$\int f \, d\mu = \lim_{n \rightarrow \infty} \int f_n \, d\mu.$$

No integrability assumption is required.

Monotone convergence: proof structure

Monotonicity immediately gives

$$\lim_n \int f_n d\mu \leq \int f d\mu.$$

For the reverse inequality, take a simple function $0 \leq S \leq f$, fix $0 < c < 1$, and consider

$$A_n = \{f_n \geq cS\} \uparrow E$$

on the support of S . Continuity from below of μ gives the required lower bound; then let $c \rightarrow 1$ and take the supremum over S .

Monotone convergence and series

For measurable functions $f_n \geq 0$,

$$S_N = \sum_{n=0}^N f_n \uparrow \sum_{n=0}^{\infty} f_n.$$

Therefore

$$\int \sum_{n=0}^{\infty} f_n d\mu = \sum_{n=0}^{\infty} \int f_n d\mu.$$

This is one of the most useful consequences of the monotone convergence theorem.

Fatou's lemma

Lemma

If the $f_n \geq 0$ are measurable, then

$$\int \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int f_n d\mu.$$

Fatou's lemma provides a one-sided inequality without any monotonicity assumption.

Set

$$g_n = \inf_{k \geq n} f_k.$$

Then $g_n \uparrow \liminf f_n$ and $g_n \leq f_k$ for every $k \geq n$. Hence,

$$\int g_n d\mu \leq \inf_{k \geq n} \int f_k d\mu.$$

Apply the monotone convergence theorem to (g_n) and then let $n \rightarrow \infty$.

Dominated convergence theorem

Lebesgue dominated convergence theorem

Assume $f_n \rightarrow f$ a.e. and that there exists $g \in L^1(\mu)$ such that

$$|f_n| \leq g \quad \text{a.e. for every } n.$$

Then $f \in L^1(\mu)$ and

$$\int f_n d\mu \rightarrow \int f d\mu.$$

Moreover,

$$\int |f_n - f| d\mu \rightarrow 0.$$

Dominated convergence: proof structure

Since $|f_n| \leq g$ and $f_n \rightarrow f$ a.e., we also have $|f| \leq g$ a.e. Apply Fatou's lemma to the non-negative sequences

$$g + f_n, \quad g - f_n.$$

This gives the lower and upper bounds forcing

$$\lim_n \int f_n d\mu = \int f d\mu.$$

How to choose the right theorem

Monotone convergence $0 \leq f_n \uparrow f$.

Fatou $f_n \geq 0$, with no domination or monotonicity assumption.

Dominated convergence $f_n \rightarrow f$ a.e. and $|f_n| \leq g \in L^1$.

Exam tip

Never write “by dominated convergence” before exhibiting an integrable dominating function.

A classical dominated-convergence example

On $[0, 1]$, let

$$f_n(x) = \frac{x^n}{1+x}.$$

Then $f_n(x) \rightarrow 0$ for every $x \in [0, 1[$ and $|f_n(x)| \leq 1$. Since $\mathbb{1}_{[0,1]} \in L^1(\lambda)$,

$$\int_0^1 \frac{x^n}{1+x} dx \longrightarrow 0.$$

Why domination is essential

On $\mathbb{R}$, let $f_n = \mathbb{1}_{[n, n+1]}$. Then $f_n(x) \rightarrow 0$ pointwise, but

$$\int_{\mathbb{R}} f_n d\lambda = 1$$

for every n .

Lesson

Pointwise convergence alone does not imply convergence of the integrals.

Continuity under the integral sign

Let $F(u) = \int_{\Omega} f(u, x) d\mu(x)$. Assume that:

- ① $x \mapsto f(u, x)$ is measurable;
- ② $u \mapsto f(u, x)$ is continuous at u_0 for almost every x ;
- ③ $|f(u, x)| \leq g(x)$ with $g \in L^1$, independently of u .

Then

$$F(u) \rightarrow F(u_0) \quad (u \rightarrow u_0).$$

Continuity: mechanism

For every sequence $u_n \rightarrow u_0$,

$$f(u_n, x) \rightarrow f(u_0, x) \quad \text{a.e.}$$

and

$$|f(u_n, x)| \leq g(x).$$

The dominated convergence theorem gives

$$F(u_n) \rightarrow F(u_0).$$

Continuity under the integral sign is therefore essentially an application of the dominated convergence theorem.

Example: Euler's Gamma function

For $s > 0$,

$$\Gamma(s) = \int_0^{\infty} x^{s-1} e^{-x} dx.$$

On a compact interval $s \in [a, b] \subset (0, \infty)$,

$$x^{s-1} e^{-x} \leq (x^{a-1} + x^{b-1}) e^{-x},$$

and the right-hand side is integrable. Hence Γ is continuous on $(0, \infty)$.

Differentiation under the integral sign

Let

$$F(u) = \int_{\Omega} f(u, x) d\mu(x).$$

Assume that $u \mapsto f(u, x)$ is differentiable a.e. and that

$$\left| \frac{\partial f}{\partial u}(u, x) \right| \leq g(x), \quad g \in L^1,$$

uniformly in u on the interval under consideration. Then

$$F'(u) = \int_{\Omega} \frac{\partial f}{\partial u}(u, x) d\mu(x).$$

Why can we differentiate?

For $u_n \rightarrow u_0$, consider the difference quotients

$$\varphi_n(x) = \frac{f(u_n, x) - f(u_0, x)}{u_n - u_0}.$$

They converge a.e. to $\partial_u f(u_0, x)$. The mean value theorem and domination of the derivative give

$$|\varphi_n(x)| \leq g(x).$$

Apply the dominated convergence theorem.

Example of differentiation under the integral sign

For $t > 0$,

$$F(t) = \int_0^{\infty} \frac{e^{-tx}}{1+x^2} dx.$$

For $t \in [a, b] \subset (0, \infty)$,

$$\left| \frac{\partial}{\partial t} \frac{e^{-tx}}{1+x^2} \right| = \frac{xe^{-tx}}{1+x^2} \leq xe^{-ax},$$

which is integrable. Therefore

$$F'(t) = - \int_0^{\infty} \frac{xe^{-tx}}{1+x^2} dx.$$

Computing a Lebesgue integral in practice

- 1 If f is simple: use $\sum_k x_k \mu(A_k)$.
- 2 For counting measure: reduce the integral to a series.
- 3 For Lebesgue measure and a usual piecewise-continuous function: use standard integration techniques.
- 4 For limits: identify monotone convergence, Fatou, or dominated convergence before manipulating the integral.

Dominated-convergence checklist

To justify

$$\int f_n d\mu \rightarrow \int f d\mu,$$

write explicitly:

- 1 $f_n \rightarrow f$ pointwise or a.e.;
- 2 each f_n is measurable;
- 3 find g independent of n such that $|f_n| \leq g$;
- 4 prouver que $g \in L^1(\mu)$;
- 5 conclude using the dominated convergence theorem.

Key takeaways

- Simple functions are the elementary building blocks.
- The integral of a non-negative function is defined by approximation from below.
- The general case uses $f = f^+ - f^-$; integrability means $\int |f| < \infty$.
- The integral is linear, monotone, and unchanged by modifications on a null set.
- Monotone convergence, Fatou's lemma, and dominated convergence control passage to the limit.
- Dominated convergence is also the main tool behind continuity and differentiation under the integral sign.

Where to practise?

Tutorial 3

The associated tutorial covers measurable functions, approximation by simple functions, the three convergence theorems, and their applications.

Next step

This first version is deliberately comprehensive. It can later be shortened to fit the available class time.